

Lab 2: Agile Backlog Creation and Sprint Simulation in Jira

Lab Equipment Reservation System (Jira project key: HARSH)

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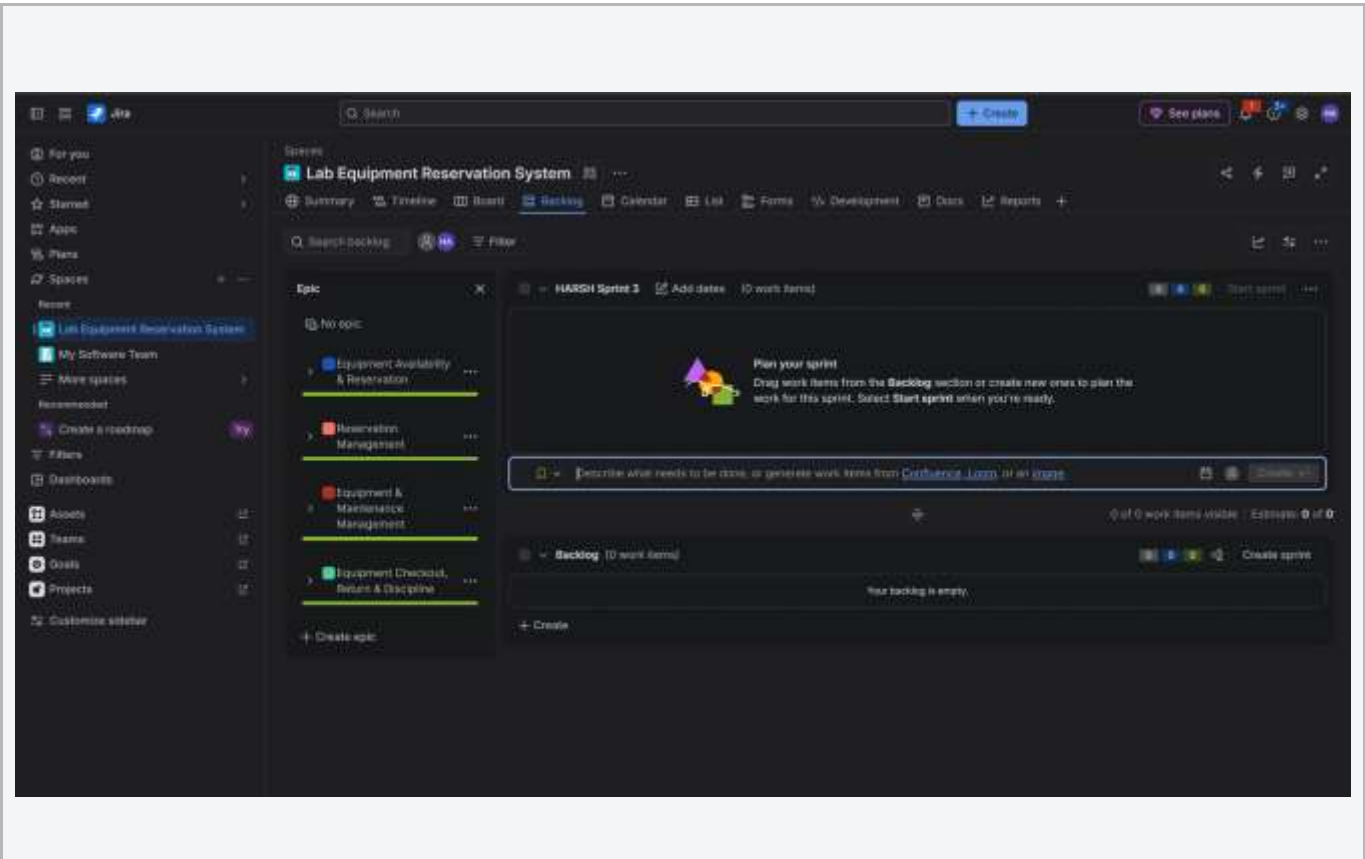
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1. Epics

The Lab Equipment Reservation System is organized into four major functional themes. Each Epic groups related user stories from the Lab 1 functional requirements.

Epic	Description	Story points
EPIC 1: Equipment Availability & Reservation	Enable students to view equipment availability and reserve calibrated lab equipment for valid time slots.	13
EPIC 2: Reservation Management	Allow students to modify or cancel existing equipment reservations.	5
EPIC 3: Equipment & Maintenance Management	Allow lab technicians to manage equipment, calibration status, maintenance and unavailability.	15
EPIC 4: Equipment Checkout, Return & Discipline	Track checkout and return activities and manage late returns, restrictions and penalties.	16

Total: 49 story points across 12 user stories.



Jira Backlog showing the four Epics in the Epic panel

2. Backlog: User Stories under their Epics

The backlog contains the 12 user stories created from the Lab 1 functional requirements. Story points use the Fibonacci scale, and priorities are based on user impact and business value.

Key	Story	User story	Priority	Pts	Sprint
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HARSH-6	1.1 View Equipment Availability	As a student, I want to view available lab equipment and time slots, so that I can choose equipment that is available when I need it.	High	3	Sprint 1
HARSH-7	1.2 Reserve Lab Equipment	As a student, I want to reserve calibrated lab equipment for a selected time slot, so that I can use the required equipment for my laboratory work.	High	5	Sprint 1
HARSH-8	1.3 Enforce Reservation Rules	As a student, I want the system to enforce the maximum 2-hour reservation duration and 7-day advance booking limit, so that equipment is allocated fairly.	High	5	Sprint 1
HARSH-9	2.1 Modify Reservation	As a student, I want to modify my existing equipment reservation, so that I can change my booking when my schedule changes.	Medium	3	Sprint 2
HARSH-10	2.2 Cancel Reservation	As a student, I want to cancel an existing equipment reservation, so that the equipment becomes available for other students when I no longer need it.	High	2	Sprint 2
HARSH-11	3.1 Manage Equipment	As a lab technician, I want to add, update, and manage laboratory equipment information, so that the system contains accurate information about available equipment.	High	5	Sprint 1
HARSH-12	3.2 Manage Calibration Status	As a lab technician, I want to record and update the calibration status of equipment, so that students can only reserve equipment that is properly calibrated.	High	5	Sprint 1
HARSH-13	3.3 Manage Maintenance & Unavailability	As a lab technician, I want to mark equipment as under maintenance or unavailable, so that students cannot reserve equipment that cannot currently be used.	High	5	Sprint 2
HARSH-14	4.1 Record Equipment Checkout	As a lab technician, I want to record when equipment is checked out to a student, so that the system maintains an accurate record of equipment currently in use.	High	3	Sprint 2
HARSH-15	4.2 Record Equipment Return	As a lab technician, I want to record when equipment is returned, so that the equipment can become available for future reservations.	High	3	Sprint 2
HARSH-16	4.3 Detect & Flag Late Returns	As a lab technician, I want the system to identify and flag late equipment returns, so that overdue equipment can be followed up appropriately.	High	5	Sprint 2
HARSH-17	4.4 Apply Restrictions or Penalties	As a lab technician, I want to apply restrictions or penalties to students with late returns, so that repeated late returns are discouraged and equipment availability is protected.	Medium	5	Sprint 2

3. User Stories, Priority and Story Points

Priority is assigned as High or Medium according to user impact. Story points estimate relative effort, complexity and uncertainty.

Priority	Number of stories	Story points	Use
High	10	41	Core reservation, equipment and lifecycle functionality
Medium	2	8	Useful management functions with lower immediate impact
Total	12	49	Complete backlog

4. Sprint Plan

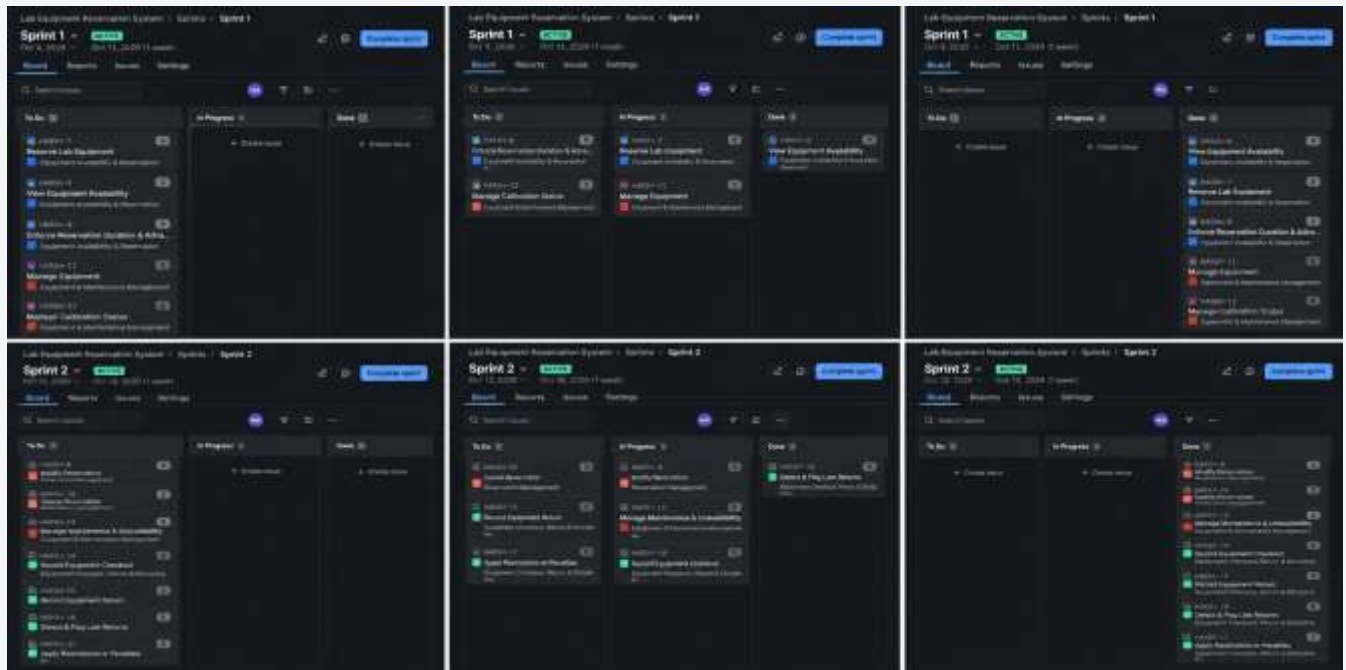
Two one-week sprints are used to simulate implementation. Sprint 1 focuses on the core reservation foundation; Sprint 2 completes reservation management and the equipment lifecycle.

Sprint	Stories	Points	Goal
Sprint 1 (1 week)	1.1, 1.2, 1.3, 3.1, 3.2	23	Students can view availability and reserve equipment while technicians manage equipment and calibration.
Sprint 2 (1 week)	2.1, 2.2, 3.3, 4.1, 4.2, 4.3, 4.4	26	Users can manage reservations and technicians can manage maintenance, checkout, return and late-return discipline.
Backlog after Sprint 2	None	0	All 12 stories are included across the two simulated sprints.

5. Sprint Board (Active Sprint View) — Sprint 1 and Sprint 2

Sprint 1 starts with five stories (23 points) in To Do. During the simulation, stories are moved through To Do → In Progress → Done.

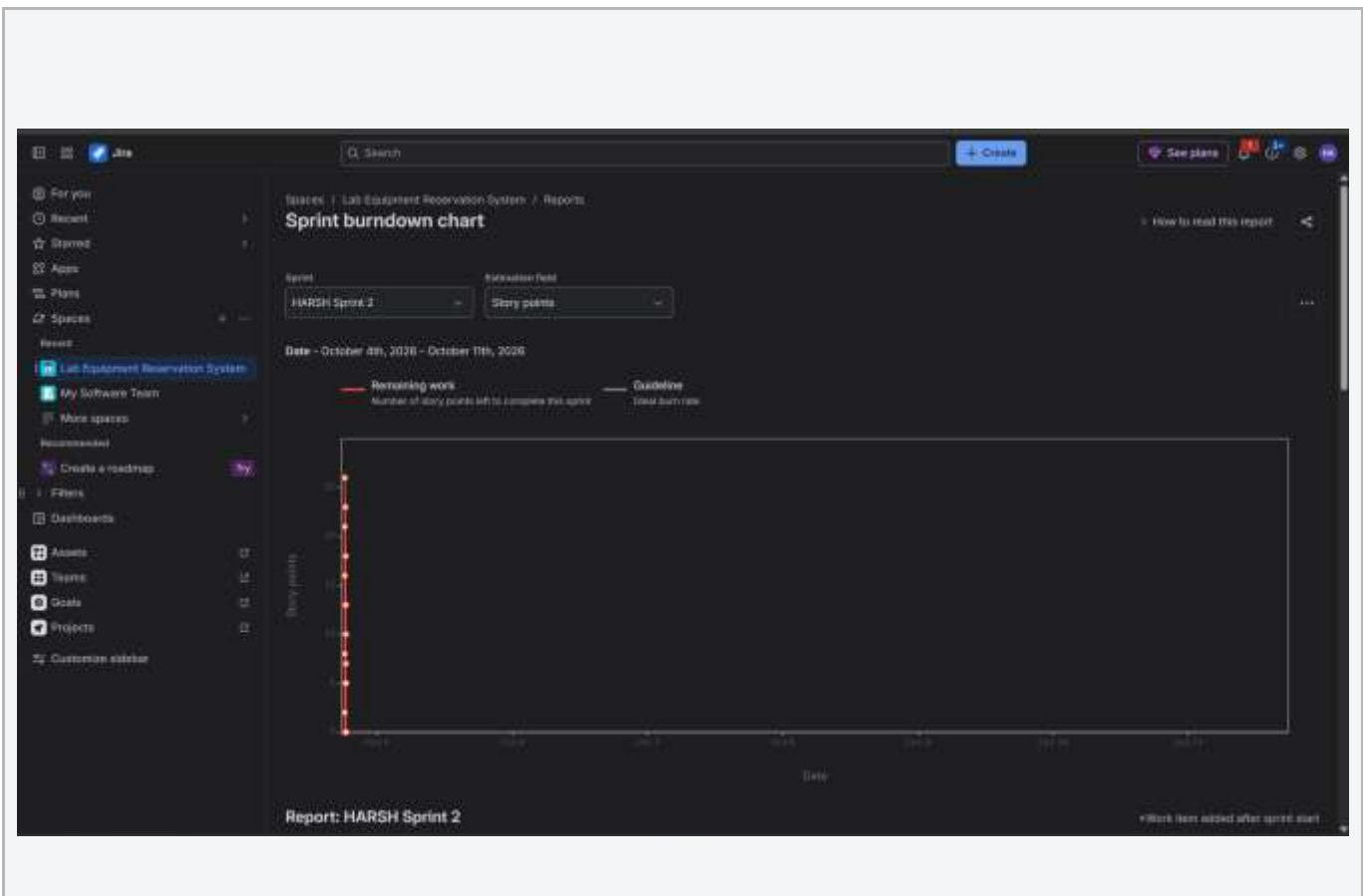
Sprint 2 contains the remaining seven stories (26 points) and is simulated in the same way.



7. Burndown Charts

The burndown chart shows remaining story points during each sprint. Because the lab uses a simulated sprint completed in a short time, the chart may show rapid drops. The chart should be captured from Jira Reports → Burndown Chart.





Sprint 2 Burndown Chart

8. Reflection

1. Did your estimations reflect the actual effort?

The estimates provide a reasonable relative comparison of effort. Smaller 2–3 point stories represent simpler reservation-management or record-keeping work, while 5-point stories involve more rules, validation, equipment state or uncertainty. Since the sprints were simulated in minutes rather than a real week, the actual time spent cannot be used to validate the estimates. The estimates are therefore useful mainly as a relative measure of complexity.

2. Was your backlog well-prioritized?

Yes. Core availability, reservation, reservation-rule enforcement, equipment management and calibration were given High priority because they are necessary for the main purpose of the system. Modify Reservation and Apply Restrictions or Penalties were given Medium priority because they support the core workflow but are not required for the first usable version.

3. How did your simulated sprint align with your plan?

Sprint 1 was planned with 23 points covering the core reservation foundation, while Sprint 2 contained the remaining 26 points for reservation management and the equipment lifecycle. The stories were selected before starting each sprint and then moved through To Do, In Progress and Done. Because the activity is a simulation, the elapsed time does not represent a real one-week development sprint.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart shows how quickly committed story points were completed. A rapid drop is expected because the sprint was simulated in minutes. The final commitment of 23 points for Sprint 1 and 26 points for Sprint 2 provides a starting reference for capacity, but a real team would need several actual sprints before using velocity as a reliable planning measure.